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MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)
REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCFTQ100 FUNDAMENTALS OF FOOD TECHNOLOGY

(2024 ADMISION ONWARDS)

Duration: 2 Hours Maximum Marks: 50 Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap) Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A Short Answer Type Questions

Answer any 5 Questions Each Question carries 2 marks

1. Write some examples of cereals, millets and pulses. [K] / [CO1]Cereals: Whole Grain Cereal, Corn-Based Cereals, Rice-Based Cereals

Millets- Examples: Pearl Millet (BajraFinger Millet (Ragi),Foxtail Millet,Proso Millet, Barnyard Millet

Pulses examples-Bengal Gram, Black Gram, Cowpea Bean, Green Gram**.**

2. Comment on gel. [K] [CO1, CO2, CO4] Gel formation is the transition of a liquid into a semi-solid state, often through the use of gelling agents.

Gelling Agents- Gelatin, Agar-Agar, Pectin,Carrageenan.

3. Write the difference between prebiotics and probiotics. [K] / [CO1, CO4]

Prebiotics	Probiotics
Are fibers/compounds	Are live microbes
Feed good bacteria	Add new good bacteria
Not alive	Living organisms
Found in high-fiber plant foods	Found in fermented foods

4. What are theexamples of enriched foods and their benefits? [K] / [CO1, CO4]Enriched foods are products that have added nutrients to enhance their nutritional value. Common examples include:

Fortified Cereal -**Benefits:** Often enriched with vitamins and minerals like iron, folic acid, and B vitamins, which support overall health and help prevent deficiencies.

Milk and Dairy Products. **Benefits:** Fortified with vitamin D and calcium, promoting bone health and reducing the risk of osteoporosis.

Bread and Flour-**Benefits:** Commonly enriched with iron and B vitamins (like thiamine and niacin), which are essential for energy metabolism and red blood cell production.

Juices**Benefits:** Orange juice is frequently fortified with calcium and vitamin D, aiding in bone health and providing a non-dairy source of these nutrients.
Plant-Based Milk Alternatives**. Benefits**: Often enriched with calcium, vitamin D, and B12, making them suitable substitutes for those who are lactose intolerant or following a vegan diet.
Infant Formula-**Benefits:** Enriched with essential fatty acids, vitamins, and minerals to support growth and development in infants.
Nutritional Yeast-**Benefits:** Often fortified with B vitamins, including B12, which is important for energy production and neurological function, especially for vegans**.**

5. What is the general principle of Codex? [U] / [CO3, CO5]The general principle of Codex revolves around the establishment of international food standards to ensure food safety, quality, and fair trade practices.
Key Objectives:Food Safety: Codex aims to protect consumer health by setting safety standards for food products.Quality Assurance: It establishes guidelines to ensure that food products meet certain quality criteria.Fair Trade Practices: Codex promotes fair practices in food trade, helping to facilitate international commerce**.**

6. Define extruded food products. [An] / [CO1]Extruded food products are items created through a process called extrusion, where ingredients are mixed, cooked, and shaped under high pressure and temperature. This method is commonly used in the production of snacks, cereals, and pet foods**.**

7. Write the advantages of BIS. [K] / [CO3, CO5BIS(Bureau of Indian Standards) plays a crucial role in standardization and quality assurance in India. Its advantages include:
Quality Assurance-Establishes standards for products and services to ensure safety and reliability.Conducts testing and certification to verify compliance with these standards.
Consumer Protection- Protects consumers by ensuring that products meet safety and quality benchmarks.Provides a mechanism for addressing consumer grievances related to product quality**. (5x2=10 marks)**

Part B Short Essay Type Questions Answer any 5 Questions Each Question carries 4 marks

8. Briefly explain about function and benefits of food technology[K] / [CO1 Function of Food Technology:Food technology focuses on the processing, preservation, packaging, storage, and distribution of food.Its main functions include: Ensuring food safety (preventing contamination and spoilage).Extending shelf life through preservation methods Improving food quality in taste, texture, and nutrition Developing new food products (e.g., plant-based meats, fortified foods. Enhancing food processing efficiency through machinery and automation Benefits of Food Technology: Safer food through proper hygiene, testing, and preservation Longer shelf life, reducing food waste, Better nutrition via fortification (adding vitamins/minerals)

9. Write a note on the properties of colloidal systems. [K] / [CO1, CO2, CO4] 1. Tyndall Effect 2. Brownian Motion 3. Adsorption 4. Electrical Charge (Electrokinetic Properties) 5. Stability 6. Coagulation or Flocculation 7. Dialysis 8. Optical Properties

10. What role does surface tension play in the texture and mouthfeel of food products, such as dressings and sauces? [K] / [CO1]Surface tension significantly influences the texture and mouthfeel of food products like dressings and sauces.

Texture and Consistency: Surface tension affects how liquids spread and coat surfaces. Higher surface tension can lead to thicker textures, while lower surface tension allows for smoother, more fluid products.Emulsifiers, such as lecithin in dressings, reduce surface tension, promoting a creamier mouthfeel and better mixing of oil and water phases.

Mouthfeel: The perception of mouthfeel is influenced by how a product interacts with the tongue and palate. Lower surface tension can create a more uniform and pleasant coating sensation.Products with high surface tension may feel more viscous or sticky, impacting overall enjoyment.

11. Write a short note on specific gravity, freezing point and humidity. [An] / [CO1, CO2CO4]

Specific gravity, freezing point, and humidity are important physical properties relevant in various scientific and practical applications.

Specific Gravity :Specific gravity is the ratio of the density of a substance to the density of a reference substance, typically water for liquids and solids.

Freezing Point: The freezing point is the temperature at which a liquid turns into a solid.

For pure water, the freezing point is 0°C (32°F) at standard atmospheric pressure..

Humidity: Humidity refers to the amount of water vapor present in the air.

It is usually expressed as a percentage, known as relative humidity, which compares the current moisture level to the maximum moisture level the air can hold at a given temperature.

12. Write about toxic effects of some metals and chemicals. [K] / [CO1, CO4]Toxic effects of metals and chemicals can lead to serious health issues. Exposure can occur through ingestion, inhalation, or skin contact.

Heavy Metals: Lead: Affects the nervous system, particularly in children, leading to developmental delays and cognitive impairments. Chronic exposure can cause hypertension and kidney damage.

Mercury: Impairs neurological function and can lead to tremors, memory problems, and cognitive deficits. Methylmercury, a toxic form, accumulates in fish and poses risks to pregnant women and developing fetuses.

Cadmium: Linked to kidney damage and bone fragility. It can also cause respiratory issues when inhaled, particularly in occupational settings.

Chemicals: Benzene: A known carcinogen that can cause blood disorders, including leukemia. Long-term exposure affects bone marrow and immune function.

Formaldehyde: Associated with respiratory issues and skin irritation. Prolonged exposure may increase cancer risk, particularly nasopharyngeal cancer.

Asbestos: Causes lung diseases, including asbestosis and mesothelioma. Inhalation of asbestos fibers leads to severe respiratory problems.

13. State significance of FSSAI in food industry. [K] / [CO3, CO5]The Food Safety and Standards Authority of India (FSSAI) plays a crucial role in ensuring food safety and quality in the food

industry.Regulatory AuthorityFSSAI establishes standards for food products and regulates their manufacture, storage, distribution, and sale.It ensures compliance with food safety laws and guidelines to protect consumer health.Consumer Protection, Quality Assurance

14. Explain the procedure of detecting adulterants using the DART method. [U] / [CO2]The DART (Direct Analysis in Real Time) method is a rapid technique used for detecting adulterants in various substances, particularly in food and pharmaceuticals.

Procedure**(5x4=20 marks)**

Part C Long Essay Type Questions Answer any 2 Questions Each Question carries 10 marks

15. Discuss in detail the types of adulterants in food, highlighting both intentional and incidental adulterants with examples.[U] / [CO2 Food adulteration involves the addition of substances that compromise the quality and safety of food. Adulterants can be classified into two main categories: intentional and incidental.

Intentional Adulterants

These are deliberately added to enhance appearance, weight, or shelf life, often at the expense of safety.Examples:Water in Milk: Diluting milk with water to increase volume, Artificial Colors: Using synthetic dyes in spices like turmeric to enhance color, Starch in Ice Cream: Adding starch to increase thickness and reduce production costs.

Lead Chromate in Turmeric: A toxic substance used to enhance color.

Incidental Adulterants: These are unintended contaminants that occur during production, processing, or storage.Examples:Pesticide Residues, Heavy Metals, Microbial Contaminants, Packaging Materials

16. Compare fabricated food and GM food.[K] / [CO1 Fabricated food and genetically modified (GM) food differ primarily in their production processes and purposes.Fabricated Food: Also known as synthetic or lab-grown food, fabricated food is created through artificial processes, often using ingredients that mimic traditional food components.Production: It typically involves techniques like fermentation, extrusion, or 3D printing to create food products that may not exist in nature.

Purpose: Aims to replicate flavors, textures, and nutritional profiles of traditional foods, often for dietary preferences or sustainability.

Genetically Modified Food: GM food is produced by altering the genetic material of plants or animals to achieve desired traits.Production: This involves techniques such as gene editing or transgenic methods to enhance characteristics like pest resistance, nutritional content, or shelf life.Purpose: Focuses on improving agricultural efficiency, crop yields, and food security.

17. Define organic food, characteristics, and advantages of organic farming and limitations of organic farming.[A] / [CO1]Organic food refers to products grown and processed without synthetic fertilizers, pesticides, genetically modified organisms (GMOs), or artificial chemicals. It emphasizes sustainable farming practices that promote ecological balance and biodiversity.

Characteristics of Organic Food:Natural Ingredients: Grown without synthetic additives.

Certification: Must meet specific standards set by regulatory bodies (e.g., USDA in the U.S.)Non-GMO: Free from genetically modified organisms.

Limitations of Organic Farming:Higher Costs, Lower Yields,Limited Availability,Shorter Shelf Life,Certification

. **18. Write an essay on any three properties of food, specific gravity, humidity, freezing point[U] / [CO1, CO2, CO4**Food properties such as specific gravity, humidity, and freezing point play crucial roles in food science and preservation.**Specific Gravity**:Specific gravity measures the density of a substance compared to the density of water.It helps in determining the composition and quality of food products.For example, oils have a lower specific gravity than water, which is why they float.This property is essential in food processing and formulation, influencing texture and mouthfeel.

Humidity: Humidity refers to the amount of moisture present in food.It affects the shelf life, texture, and flavor of food products.High humidity can lead to spoilage and mold growth, while low humidity can cause dehydration and loss of quality.

Freezing Point:The freezing point is the temperature at which a liquid turns into a solid. Different foods have varying freezing points based on their composition, such as sugar and salt content.

(2x10=20 marks)